

UNIVERSITY OF MAURITIUS

FACULTY OF INFORMATION, COMMUNICATION AND DIGITAL TECHNOLOGIES



SECOND SEMESTER EXAMINATIONS

AUGUST/SEPTEMBER 2022

PROGRAMME	BSc (Hons) Information Systems-Level 3 (IC311/19)		
MODULE NAME	Information Systems Security		
DATE	Wednesday 17 August 2022	MODULE CODE	SIS3118Y
TIME	09.45-11.45 hrs	DURATION	2 Hours
NO. OF QUESTIONS SET	4	NO. OF QUESTIONS TO BE ATTEMPTED	4

INSTRUCTIONS TO CANDIDATES

This question paper consists of **Section A** and **Section B**.

Answer all questions from each Section.

Use separate answer books for each Section.

Question 1

- (a) (i) A organisation wishes to set up a comprehensive security plan for its IS infrastructure. Identify the elements that must be included in the plan. **[3 marks]**
- (ii) Discuss the statement “An organisation’s security policy can be properly enforced only if accountability is maintained”. **[3 marks]**
- (b) Explain how the concept of layering simplifies security. **[2 marks]**
- (c) The simplest way to implement an access control policy is through the use of an Access Control Matrix (ACM). The following shows the content of an ACM.

	F1	F2	F3	F4
Domain 1	Read, Write	Execute	Read	
Domain 2		Write		Print
Domain 3	Execute		Read	Print

- (i) Determine whether a user in Domain 1 is allowed to execute object, F1. **[2 marks]**
- (ii) Determine whether a user in Domain 3 is allowed to read object, F3. **[2 marks]**
- (iii) Represent the matrix as a capability list. **[6 marks]**
- (iv) What capabilities would be given to a user in Domain 2? **[2 marks]**
- (d) (i) A program is written to compute the sum of the integers from 1 to 10. The programmer, well trained in reusability and maintainability, writes the program so that it computes the sum of the numbers from i to n.

However, a team of security specialists scrutinizes the code. The team certifies that this program properly sets i to 1 and n to 10; therefore, the program is certified as being properly restricted in that it always operates on precisely the range 1 to 10.

List three different ways that this program can be attacked so that during execution it computes a different sum, for example, 4 to 13.

[3 marks]

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- (ii) Cipher Feedback Mode (CFB) is a block cipher that may allow data to be encrypted in units smaller than the block size. Explain how the encryption process may take place.

[2 marks]

(Total: 25 Marks)

Question 2

- (a) Kerberos is used as an authentication method that is used by many operating systems including Windows. Generally, Kerberos does not automatically trust clients. Instead, it requires a client to prove its identity through a third party. Describe fully how the authentication is performed.

[5 marks]

- (b) (i) Password Authentication Protocol (PAP) is the most basic form of authentication. Describe how the PAP functions.

[3 marks]

- (ii) A hacker makes use of password guessing to access a system. If the passwords are six uppercase alphabetic characters long, how much time would it take to determine a particular password, assuming that testing an individual password requires 0.002 seconds? Show all your workings.

[5 marks]

- (c) (i) Feistel Cipher is a symmetric cipher used in the construction of block ciphers. State three advantages of using Feistel ciphers.

[3 marks]

- (ii) Digital signatures make use of Personal Key Infrastructure (PKI) in the signing process. Describe how the authentication is made when Alice sends a message to Bob using a digital signature.

[5 marks]

- (d) An organisation decided to use biometric authentication to control access to employees and registered visitors only. The system, however, produced some false negatives i.e. some genuine employees were falsely denied access.

- (i) Discuss how the organisation can overcome this problem.

[2 marks]

- (ii) State the security advantage that justifies the extra expense on biometric devices despite the resulting false-negative cases.

[2 marks]

(Total: 25 Marks)

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SECTION B

Question 3

- (a) There are three models of intrusion detection mechanisms: anomaly-based detection, signature-based detection, and hybrid detection.
- (i) Discuss the difference between anomaly-based detection and signature-based detection.
[6 marks]
- (ii) The anomaly-based detection and signature-based detection have a number of drawbacks. List two(2) drawbacks for each.
[2+2 marks]
- (b) Intrusion detection systems are classified based on their monitoring scope. There are network-based intrusion detection and host-based detections. Discuss the difference between network-based intrusion detection and host-based detections.
[6 marks]
- (c) Explain the term “Honeypots” as used in Information Systems Security.
[4 marks]
- (d) Discuss the functions of an Incident Response Team.
[5 marks]
(Total: 25 Marks)

Question 4

- (a) Define the term Digital Forensic Science and list the purpose of the science.
[3 + 3 marks]
- (b) Distinguish between Cryptography and Digital Watermarking.
[3 + 3 marks]
- (c) User identity can be protected by using Privacy-Enhancing Technologies (PETs) such as Anonymity, Pseudonymity, Unobservability and Unlinkability. Define each of these terms and explain how they are used for privacy enhancement.
[4x2 marks]
- (d) What is Bluetooth? Discuss the security issues related to the use of Bluetooth technology.
[5 marks]
(Total: 25 Marks)

END OF QUESTION PAPER